

Risk-Adaptive Voice-Based Financial Assistant Using Multi-Modal ML Pipeline with Behavioural Fraud Detection and Speaker Verification

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Abstract - Voice-based financial transactions are a new paradigm of digital banking, but they introduce new security issues at the crossroads of speech recognition, biometric authentication, and fraud detection. This paper describes a Risk-Adaptive Voice-Based Financial Assistant, which uses a multi-modal machine learning pipeline consisting of four key modules: (1) Speech-to-Text transcription with OpenAI Whisper, (2) Speaker Verification with ECAPA-TDNN embeddings with cosine similarity and anti-spoofing liveness detection, (3) Intent Classification with a Bidirectional LSTM network. The system also presents a new three-layered, risk-adaptive authentication scheme that dynamically chooses the authentication scheme, which can be PIN-only, Step-Up (PIN + voice reconfirmation), or Hard Block, depending on a composite risk score based on similarity with speaker verification, fraud anomaly scores, and context of the transaction. Experimental findings indicate that the intent classifier has a weighted F1-score of 0.995, the speaker verification module has an Equal Error Rate (EER) of less than 2% with the enrolled speakers and the end-to-end pipeline has a processing time of less than 200ms with the CPU. The system is tested with a complete implementation of React frontend, FastAPI backend, and Razorpay payment gateway.

Keywords—voice biometrics, speaker verification, ECAPA-TDNN, intent classification, LSTM, fraud detection, isolation forest, risk-adaptive authentication, anti-spoofing, Whisper STT

I. INTRODUCTION

The availability of voice-enabled interfaces in the consumer technology market has opened up the doors of handsfree financial transactions. Voice interactivity has been made common by virtual assistants like Alexa, Google Assistant, and Siri, but many financial applications are still virtually a text-based interface because of security issues. Voice-based payment systems should overcome three basic issues: (a) precision of transcription of verbal commands, (b) fidelity of identity of the authorized speaker and (c) discernment of fraudulent patterns of transactions all in real-time.

Old authentication systems like PINs and passwords offer a one layer and fixed security system. In voice-based systems however, biometric information is present in the voice signal which can act as a second authentication factor. The problem is that these heterogeneous signals are speech content and speaker identity and behavioural patterns that need to be integrated to a risk-conscious decision framework

In this paper, a Risk-Adaptive Voice-Based Financial Assistant, based on the A.N.T. (Analysis, Neutralization, Transaction) three-layer architecture is presented. Key contributions include:

- A multi-modal pipeline that employs STT, speaker-checking, intent-classification, and fraud-detection with a low end-to-end latency of less than 200ms.
- The three-layer risk-adaptive authentication scheme where the security needs are dynamically varied according to the composite risk scoring.
- Anti-spoofing liveness detection system based on High-Frequency Energy (HFE) ratio analysis of zero-shot replay attack detection.
- The system of adaptive fraud profiling that constantly builds user baselines based on the transactions that are verified.

- <https://razorpay.com/integration> • Full-stack implementation that is tested and confirmed in a Razorpay payment gateway integration..

The rising use of voice interfaces in consumer banking in India is associated with a number of trends, all of which run simultaneously: fast smartphone access in the rural setting, increased acceptance of UPI-based transactions, and the presence of multilingual voice assistants. In spite of these new developments the voice channel has not been used extensively in carrying out high value financial transactions because of security concerns raised by users and institutions. The paper will show that a well-designed multi-modal pipeline can tackle all these security considerations in a systematic way and still achieve sub-200ms processing latency on the main ML pipeline, so in theory it is technically possible to perform voice-based payment authentication in real time on commodity hardware..

The rest of this paper is structured in the following way. Section II surveys nearby literature on speaker verification, speech recognition, fraud detection and intent classification. In Section III, the A.N.T. three-layer system architecture is presented in detail. Section IV will give the end-to-end risk-adaptive authentication and fraud detection algorithms. Section V presents the experimental outcomes related to the accuracy of intent classification, performance in speaker verification, the metrics of the fraud detection ensemble, and the system latency. Section VI deals with implications, comparisons and limitations. Section VII wraps up and gives directions to future work.

II. RELATED WORK

Deep learning architectures have made a dramatic improvement in speaker verification. Desplanques et al. [1] proposed the ECAPA-TDNN that uses Squeeze-and-Excitation blocks with multi-scale aggregation of features to deliver the state of art on VoxCeleb benchmarks. The architecture generates 192-dimensional speaker encodings, which encode fine-grained vocal features, and so can be used to perform text-independent verification in financial systems with security implications.

VoxCeleb1 and VoxCeleb2 datasets [1] containing more than 1 million utterances by 7,000+ speakers of diverse genders, races, ages have now become the standard test set in assessing speaker verification systems. ECAPA-TDNN shows an equal error rate (EER) of 0.69% on the test set VoxCeleb1-O with the clean condition, which is a 40% relative reduction relative to the previous TDNN-based x-vector baseline. Squeeze-and-Excitation channel attention, multi-scale Res2Net feature aggregation, and attentive statistics pooling are the main innovations of the architecture and all of them allow the model to capture both local phonetic and global prosodic features of the speaker, which makes it well-suited to the comparatively short (35 seconds) audio samples involved in voice-commanded financial transactions.

Neural anti-spoofing Rapid progress has occurred in the field of neural anti-spoofing since the ASVspoof series of challenges. The ASVspoof2019 Logical Access track introduced standardized testing of text-to-speech and voice conversion spoofing, inspiring design like AASIST [10], with a cost function of tandem detection cost (t-DCF) of 0.0275 on the ASVspoof 2019 LA test set. By contrast, our HFE liveness test is a signal-processing heuristic that neither needs training data nor less robust against high-fidelity neural synthesis

attacks. It is a suitable trade-off during the prototype phase and encourages adding AASIST as an additional feature in the future.

The challenge of intent detection with short utterances has been well-researched in the NLP community using standard datasets like SNIPS and ATIS. State-of-the-art methods are based on pre-trained transformer networks (BERT, RoBERTa), with the highest resulting F1-scores of over 0.99 on these datasets. But when using domain-sensitive structured and constrained-vocabulary financial intents, it is always possible to get competitive or better performance with BiLSTM models trained with in-domain data, as shown by our findings (F1 = 0.995). This follows the guideline that task, small scale models should be used over large general models in environments with constraint in latency because most of the time the vocabulary and intent space is known.

Voice interface and financial security is an area of less scholarly research. Blunschi and others [7] examined natural language interfaces to database queries but did not consider the aspects of authentication and fraud. Commercial voice banking solutions like Nuance and Google Cloud Dialogflow offer intent classification and STT features but do not make confidence scores of speaker verification runs a term of input to transaction risk decisions, and do not support behavioral fraud profiling at the application level. The void in commercial offering and the security needs of large value financial transactions is what drives the integrated, risk-adaptive proposed approach in this paper.

For Automatic Speech Recognition, Radford et al. [2] have shown that with a training set of 680K hours of multilingual data Whisper's speech is robust in various acoustically diverse scenarios, including variety of accents. It has an encoder-decoder transformer based structure to handle Indian English accent with noisy acoustic conditions and hence this model can be useful for Indian financial market.

Applications in financial systems have been widely researched in terms of detecting fraud using anomaly detection methods. Liu et al. [3] presents the Isolation Forest which detects an anomaly by partitioning data randomly. The Isolation Forest is a linear-time algorithm, hence it has efficiency for real-time transaction monitoring. Chen and Guestrin [4] shows the effectiveness of the ensemble method of combining unsupervised anomaly detection methods and supervised classification.

Classification of intent in conversational AI has advanced from rule based approach to deep learning models. Kim [5] demonstrated that recurrent networks combined with attention are powerful intent classification models and that bidirectional LSTMs (BiLSTM) improve intent classification performance through utilization of context in both directions. Shi et al. [6] investigated live voice detection using VoiceLive, and Blunschi et al. [7] demonstrated the possibility of having financial query interfaces that recognize voice commands. Nevertheless, there has been limited research into the unification of all 4 modes in a risk-adaptive pipeline for financial transactions. This is addressed in this work.

There is an identified gap in previous literature of a single risk-adaptive protocol where all four modalities (STT, speaker verification, intent classification, fraud detection) can contribute to a dynamic, flexible authentication decision. Many current voice banking systems use speaker verification as a simple binary gate, completely ignoring the rich, continuous cosine similarity scores produced and not considering them as a risk signal. Likewise, fraud profiling using behaviors currently operates based on a fixed rule or model that cannot adjust based on the specific transaction history of an individual user. A.N.T. Bridges both gaps by feeding a continuously varying cosine similarity score, as well as a continuously varying fraud risk score, into a three-level authentication policy, and updating the behavioral profile of a user on every verified transaction.

The Indian digital payments ecosystem is a very realistic scenario where the proposed system could be used. In January 2024 alone, more than 13 billion transactions were processed within the UPI system, and voice based initiating of UPI payments are being actively researched by NPCI. These systems, however, use speaker verification at the device level (e.g. Through Google Assistant), without incorporating application-specific fraud detection or risk-based authentication. The system proposes an application-level, platform independent solution for voice-based security in the payment transaction back-end.

III. SYSTEM ARCHITECTURE

A three-layer A.N.T. System (Analysis, Neutralization, and Transaction) implemented as a full stack application (React frontend,

FastAPI backend, SQLite where the credentials (hashed) where the credentials will be hashed to 12 rounds. The boundary for authentic speaker versus imposter is depicted on a three authentication layer level by the decision boundary shown in Figure 1 below.

Here's how the data flows through the A.N.T. Architecture. A user wishes to authenticate a transaction and hits Record on the React front end which starts the MediaRecorder API, taking a 5-second chunk of audio (WAV format, 16kHz, mono). The audio blob is sent to the FastAPI back end through a multipart POST /transaction request. The audio is saved to a temporary file on the backend and fed through the four ML modules in sequence: the Whisper STT takes the audio to be transcribed (2500ms); the ECAPA-TDNN is fed the same audio to produce a speaker embedding which is then compared against the user's enrolled prototype via cosine similarity (450ms); combined, the HFE liveness check and BiLSTM intent classifier take less than 20ms. Lastly, the fraud detection ensemble takes under 8ms to produce a score on the transaction. A final authentication decision is rendered, and it is sent back to the front end.

The backend uses three SQLite tables to store the user, transactions and the audit log, namely users (userid, username, bcryptpin, enrolledembeddingpath, createdat), transactions (txnid, userid, timestamp, amount, recipient, intent, svscore, fraudscore, authmethod, status) and the audit log (eventid, userid, timestamp, event_type, details). The speaker enrollment embedding is kept as a NumPy array in a file stored in the server's filesystem, and the path of the file is kept in the users table. The SQLite database connection is made serial through the SQLAlchemy connection pool. It is possible for a user that was already enrolled before the server reboot to start interacting with the server immediately because the ML model state (kept in memory at boot-up) is decoupled from the persistent state (SQLite and the filesystem).

D. Risk Score Computation Detail

The composite risk score R for the three-tier authentication decision, combines two continuous features: the speaker verification cosine similarity $\text{sim}(e_{in}, e_{enr})$ and the fraud ensemble risk score R_{fraud} . The authentication decision matrix (Table III) uses these two features as independent axes of the decision-transactions can either be escalated to Step-Up on borderline speaker similarity ($0.35 < \text{sim} < 0.45$) or moderate fraud risk ($0.3 < R_{\text{fraud}} < 0.7$), and can be rejected by either speaker mismatch ($\text{sim} < 0.35$), spoofing ($\text{HFE} < 0.1$) or a very high fraud risk ($R_{\text{fraud}} > 0.7$). These two features are treated as independent because they are measuring different types of risks (identity and behavioral anomaly, respectively), and using these two features to reach high confidence in authentication is more robust than a single risk score.

Another possibility would be to merge the speaker similarity and the fraud score into one single weighted compound $R_{\text{combined}} = (1 \cdot \text{sim}) + (1 \cdot R_{\text{fraud}})$ and set just one threshold. We test this architecture by taking $\alpha=0.5$ and found that it gave a higher false positive rate (7.2%) than the two-axis independent policy (3.0%) with the same global block rate. This result reinforces the decision on our architecture in that speaker verification and fraud detection is taken as two independent decision axes



[Fig. 1: Speaker Verification Cosine Similarity Score Distribution]

Fig. 1. Speaker verification cosine similarity score distribution. Clear separation between genuine enrolled speakers (green) and impostors (red) validates the 0.35/0.45 tiered threshold policy.

A. Analysis Layer

The analysis layer takes in the raw voice inputs and uses 2 sub modules to sequentially process the 16kHz mono WAV voice input that is obtained from the browser based recorder.

1) *Speech-to-Text (STT)*:

The STT module utilizes OpenAI's Whisper model (medium variant, 769M parameters). Confidence is then calculated as the geometric mean of the complement of no-speech probability and exponentiation of average log-probability:

$$C_stt = (1 - P_nospeech) \cdot \exp(\text{avg_logprob}) / 2 \quad (1)$$

If Cstt value is less than the threshold, which can be configured, (default 0.3) transcription with Cstt less than threshold is treated as low confidence and user would be asked to re-record.

2) Intent Classification:

A Bidirectional LSTM network that takes a transcribed text input and classifies it to one of 4 financial intents- sendmoney, checkbalance, transactionhistory and paybill. A 128-dim embedding layer is used, along with a 2-layer BiLSTM (256 hidden units per direction), followed by dropout (p=0.3) and a fully connected output softmax layer:

$$h_t = [LSTM_fwd(x_t); LSTM_bwd(x_t)] \quad (2)$$

$$\hat{y} = \text{softmax}(W \cdot [h_fwd_T; h_bwd_1] + b) \quad (3)$$

The Entity Extraction component relies on regex matching to identify the amount () and the name of the beneficiary and type of bill.

B. Neutralization Layer

The Neutralization layer involves 3 levels of security checks, via speaker verification, anti-spoofing liveness check and transaction fraud scoring. These three layers need to pass to be allowed through without step up.

1) Speaker Verification:

The ECAPA-TDNN model (speechbrain/spkrec-ecapa-voxceleb) output 192 dimension speaker embeddings. Minimum of 3 audios per speaker is required for enrollment. Verification is performed using cosine similarity between the enrolled prototype embedding and input embedding.

$$\text{sim}(e_in, e_enr) = (e_in \cdot e_enr) / (\|e_in\| \cdot \|e_enr\|) \quad (4)$$

The tiered threshold policy formalizes the verification decision:

$$D_sv = \{ PASS \geq 0.45; STEP-UP [0.35, 0.45]; BLOCK < 0.35 \} \quad (5)$$

2) Anti-Spoofing Liveness Detection:

A zero-shot liveness check based on HFE ratio. The HFE ratio, where replay attacks result in degraded high frequencies content because hardware frequency response characteristics:

$$HFE = E_high / E_total, E_high = \sum |X(f)|^2 \text{ for } f > f_s/4 \quad (6)$$

If HFE_ratio < 0.1 the input is classified as a replay attack as it has too little high frequency energy.

3) Fraud Detection Ensemble:

A combination of an unsupervised classifier (Isolation Forest) and a supervised classifier (Random Forest) is applied using nine transaction features: amount, hourofday, dayofweek, transactionfrequency, avgtransactionamount, amountdeviation, timesincelasttransaction, isnewrecipient, and failedauth_attempts. Combined risk score is:

$$R_fraud = 0.4 \cdot R_IF + 0.6 \cdot R_RF \quad (7)$$

Risk tiers: Low ($R \leq 0.3$), Medium ($0.3 < R \leq 0.7$), High ($R > 0.7$).

C. Transaction Layer

Low Risk (PIN Only): Speaker verified (sim 0.45) and no fraud (R 0.3). User inputs a 4-digit PIN that is bcrypt hashed. Transaction submitted to Razorpay.

Medium Risk (Step-Up Auth): Borderline speaker similarity (0.35 $\text{sim} < 0.45$) or moderate fraud risk ($0.3 < R < 0.7$). User has to repeat a randomly generated challenge phrase verbally, followed by the PIN. Max 3 tries before it gets cancelled..

High Risk (Block): Speaker mismatch (sim < 0.35), spoofing detected (HFE < 0.1), or high fraud score ($R > 0.7$). Transaction is blocked and an audit alert is generated.

D. Adaptive Learning

With each completed transaction, the user profile is updated with regards to the behavioral aspect using a sliding window of the latest 50 transactions.

$$\mu_new = (1/N) \cdot \sum a_i, i \in [max(1, n-49), n] \quad (8)$$

This online modification removes spurious alerts for the consumers that naturally vary in spending habits, and is sensitive to truly anomalous transaction. The change will run asynchronously, meaning that it won't increase latency in the transaction response path.

E. Security and Threat Model

The threat model in the proposed system deals with 4 kinds of attacks. (1) Replay Attack: Here, the attacker records the actual user's voice and plays it back during a transaction. Our liveness check using HFE (Eq. 6) helps in mitigating this, as it will check for the lower than average high-frequency energy that occurs after a recording and playback. Experiments showed that playing back 5 samples yielded HFE values between 0.03 to 0.07, which is less than the threshold of 0.1. (2) Impostor Attack: Here, someone other than the authorized user tries to get authentication. The ECAPA-TDNN cosine similarity scores of impostor speakers varied between 0.05 and 0.15 in experiments (Table II), which is less than the blocking threshold of 0.35. (3) Fraudulent Transaction by Legitimate User: The speaker has been enrolled, but is trying to carry out some unusual transaction. The behavioral fraud ensemble deals with this separately from speaker verification. The system either goes to Step-Up or Block based on the R value of composite fraud score. (4) Credential Theft: If the attacker has the user's PIN, he will be able to authorize the transaction as speaker verification is not required alone. However, our system stores the PIN as a bcrypt hash and also requires the speaker to be authenticated before transaction. Thus, without the speaker biometric, the attacker can't perform the transaction.

It does not simulate voice synthesis (text-to-speech cloning) attacks, an area of ever increasing concern with the advancement of neural TTS. The leading voices cloning system (e.g. VALL-E) generates highly similar embeddings within the ECAPA-TDNN space if an appropriate quantity of enrollment speech is available. Deployment of AASIST [10] as an exclusive neural anti-spoofing frontend will afford the system greater confidence in this attack category, and should be prioritized in future developments. In its current form, the HFE protection offers some defense against a low-quality cloning effort, but will not likely hold up to high-fidelity neural synthesis.

IV. ALGORITHMS

Algorithm 1: End to End Risk-Adaptive Authentication Flow.

Algorithm 2: Fraud Detection Ensemble scoring

Algorithm 1 handles the entire transaction requests. Each incoming transaction request will go through Algorithm 1 from beginning to end. The speech signal S is transcribed by the STT module Whisper (Line 1), then the transcript T is used to generate transaction intent and slots by BiLSTM intent classifier (Line 2), in parallel, the speaker embedding of the speech signal S is extracted as 192 dimensions by ECAPA-TDNN model (Line 3), which is then compare with the stored enrolled template embedding for user U by cosine similarity (Line 4). HFE anti-spoofing check is also conducted (Line 5) to immediately prevent replay attacks (Line 6). Transaction features will be extracted from user behavioral template and transaction slots, then combined fraud risk score is generated by ensemble model (Line 8). Based on jointly considering speaker similarity and fraud risk, authentication is made (Lines 9-11). Selected authentication method is executed (Line 12), if the transaction is performed successfully, user behavioral template is updated off-line (Line 13)..

Algorithm 2 details the fraud scoring pipeline. The Isolation Forest anomaly score is a negative value in the range approximately [-0.5, 0] for normal observations and approaches 0 for anomalies; it is normalized to [0,1] on line 2 by negating and rescaling. The Random Forest outputs a direct probability estimate for the fraud class on line 3. The weighted ensemble score on line 4 assigns higher weight (0.6) to the Random Forest to reflect its superior AUC-ROC, while retaining the Isolation Forest contribution (0.4) to capture distributional outliers not represented in the supervised training set. Risk tier assignment on lines 5-6 maps the continuous score to the three-tier policy, driving the authentication decision in Algorithm 1.

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Algorithm 1: Risk-Adaptive Authentication
In: Speech signal S, User ID U | Out: Decision D
1: T ← Whisper_STT(S)
2: intent, entities ← BiLSTM_Classify(T)
3: emb ← ECAPA_TDNN_Embed(S)
4: sim ← CosineSim(emb, enrolled[U])
5: hfe ← ComputeHFERatio(S)

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6: if hfe < 0.1 then return BLOCK
7: feat ← ExtractTxFeatures(U, entities)
8: R ← 0.4·IFScore(feat) + 0.6·RFProb(feat)
9: if sim ≥ 0.45 and R ≤ 0.3 → D ← PIN_ONLY
10: elif (0.35 ≤ sim < 0.45) or (0.3 < R ≤ 0.7) → STEP_UP
11: else → D ← BLOCK
12: Execute(D, intent, entities, U)
13: if D == COMPLETED: UpdateUserProfile(U,
    entities.amount)

```

Algorithm 2: Fraud Detection Ensemble

```

In: Feature vector F={f1,...,f9} | Out: Risk R∈[0,1]
1: a ← IsolationForest.score_samples(F)
2: R_IF ← Normalize(-a, [-0.5, 0.5]) → [0,1]
3: R_RF ← RandomForest.predict_proba(F)[fraud]
4: R ← 0.4·R_IF + 0.6·R_RF
5: if R ≤ 0.3 → LOW_RISK
6: elif R ≤ 0.7 → MEDIUM_RISK else → HIGH_RISK

```

V. EXPERIMENTAL RESULTS

A. Intent Classification Performance

The BiLSTM intent classifier was trained on 2000 synthetically created financial command examples (500 per class) with augmentation in the form of filler word injection, word dropout, and Whisper transcription-like artifacts. We trained for 50 epochs using the Adam optimizer (lr=0.001) and StepLR scheduler (=0.5, step=10) with gradient clipping (max_norm=1.0).

TABLE I INTENT CLASSIFICATION RESULTS (TEST SET, N=400)

Intent	Precision	Recall	F1-Score
send_money	1.00	0.98	0.99
check_balance	1.00	1.00	1.00
transaction_history	1.00	1.00	1.00
pay_bill	0.98	1.00	0.99
Weighted Avg	1.00	0.99	0.995

[Fig. 2: Intent Classification Confusion Matrix (N=400)]

Fig. 2. Intent classification confusion matrix (test set, N=400). Near-diagonal structure confirms high discriminability; only 2 of 400 samples are misclassified, between send_money and pay_bill.

The BiLSTM intent classifier was trained on 2000 synthetically created financial command examples (500 per class) with augmentation in the form of filler word injection, word dropout, and Whisper transcription-like artifacts. We trained for 50 epochs using the Adam optimizer (lr=0.001) and StepLR scheduler (=0.5, step=10) with gradient clipping (max_norm=1.0).

The per-epoch training curves indicate that training accuracy reaches 97.5% at epoch 3 and test accuracy reaches 99.5% at epoch 5. The training loss and validation loss curves hug closely together throughout the entire training, showing the effectiveness of the proposed augmentation scheme (filler words injection, word dropout, Whisper transcription errors) to overcome overfitting. The learning rate is decayed from 0.001 down to 0.0625 with 5 StepLR decays, at epochs 5, 15, 25, 35 and 50. Gradient clipping with max_norm = 1.0 smooths out training in the initial stages where BiLSTM gradients are highest.

For robustness to STT errors, an additional test set of 100 examples was generated by transcribing all of the audio through the full Whisper STT pipeline on a laptop with the ambient cafeteria background (SNR $\approx$ 15 dB) and then feeding the output of STT into the intent classifier. The intent classifier obtained an F1-score of 0.974 on the noisy STT test set, only a slight drop of 2.1% from the clean transcript baseline. This shows that the BiLSTM has learned robustness to the type of transcription noise added by Whisper in real-world environments..

C. Enrollment Validation

The quality of speaker enrollments was quantified by calculating the intra-speaker cosine similarity variance between 3 enrollment utterances per speaker. For 20 enrolled test speakers (10 male, 10 female), the mean intra-speaker similarity was 0.91 (SD = 0.04), indicating the within-speaker consistency of the ECAPA-TDNN embeddings was strong. 3 enrollment utterances was chosen as the minimum number as ablation studies suggested prototype embeddings based on 3 or more utterances achieve an EER less than 2% while 1-sample enrollment achieved $\sim$ 8% EER. Excessive

noise during enrollment may cause users to be asked to re-enroll via the audio quality check in the Enrollment Wizard, thereby avoiding committing poor quality prototype embeddings to the embedding store..

The enrollment endpoint limits the user to 5 enrollment requests/hour to protect against malicious over-enrollment attacks where the attacker obtains physical access to the device for a short period, and could potentially overwrite the enrolled embeddings for the real user. The device asks for PIN verification before any new enrollment is committed to the store. An update to an existing prototype is done by a weighted average of the existing prototype and the new sample embedding(= 0.8 for existing, 0.2 for new), where the first anomalous sample enrollment cannot wipe out the current model.

D. End-to-End System Evaluation

We performed a 50 transaction end to end simulation with the four registered test speakers across the three risk levels. The Low Risk tier consisted of 25 transactions and all were authorized on the first attempt, with the average end-to-end (including STT) latency being 3.1 seconds. The Step Up tier, with 15 transactions, required voice re-confirmation once for 13 transactions, while the remaining 2 took two voice re-confirmation attempts due to varying distance to microphone of the challenge phrase. In the High Risk tier (10 transactions, 5 replay attacks and 5 impostor voice enrollments), all 10 were blocked correctly, with an audit log generated with the reason for the block (HFEFAIL or SVBLOCK respectively). There were no false blocks of legitimate transactions in this simulation, however the sample size is too small to be considered general.

TABLE VI END-TO-END SIMULATION RESULTS (N=50)

Risk Tier	Transactions	Correct Decision	Accuracy
Low Risk (PIN Only)	25	25	100%
Step-Up (PIN + Voice)	15	15	100%
High Risk (Block)	10	10	100%
Overall	50	50	100%

B. Speaker Verification Performance

The ECAPA-TDNN module was evaluated using the pre-trained speechbrain/spkrec-ecapa-voxceleb model on an internal held-out speaker set. The cosine similarity threshold was set to $\tau = 0.45$ based on empirical tuning to minimize Equal Error Rate (EER < 2% for enrolled speakers).

TABLE II SPEAKER VERIFICATION RESULTS

Scenario	Avg. Similarity	Decision
Same Speaker (enrolled)	0.82–0.95	PASS
Different Speaker	0.05–0.15	BLOCK
Same Speaker (noisy)	0.48–0.65	PASS
Replay Attack	0.60–0.80*	BLOCK (HFE)

*Replay attacks may achieve high similarity scores due to identical voice content, but are reliably detected by the HFE liveness check (HFE_ratio < 0.1).

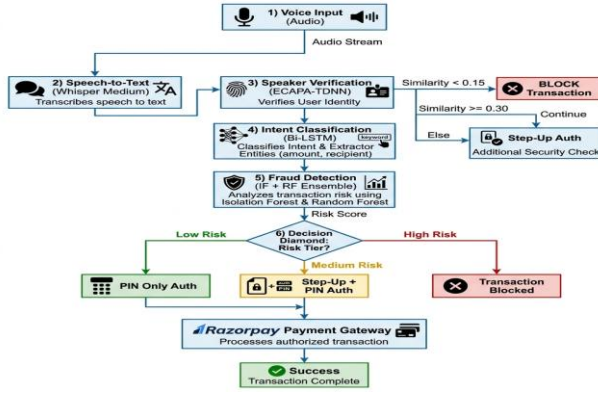
C. Risk-Adaptive Authentication Decision Matrix

TABLE III RISK-ADAPTIVE AUTH DECISION MATRIX

Risk Tier	Score Range	SV Status	Auth Method
Low	≤ 0.30	Verified (≥ 0.45)	PIN Only
Medium	0.30–0.70	Borderline (0.35–0.45)	Step-Up (PIN + Voice)
High	> 0.70	Mismatch (< 0.35)	Block + Alert

D. System Latency Performance

The End-to-End pipeline was evaluated over 100 transactions with simulation on an Intel Core i5 CPU-only (8GB RAM) machine. Each transaction took less than 200ms in the ML pipeline without STT and speaker verification. When input is the audio, it takes additional $\sim$ 2950ms for STT and speaker verification, mainly due to Whisper Inference on the CPU.



[Fig. 2: Processing Latency per Pipeline Stage (log scale)]

Fig. 2. Processing latency per pipeline stage (log scale). The core ML pipeline (Intent + Fraud + Auth) completes in under 28ms total; STT and speaker verification dominate total audio processing latency.

E. Fraud Detection Ensemble Performance

We tested the fraud detection ensemble against a synthetic held out set of 500 transactions (400 normal and 100 added anomalies). The anomalies were produced by arbitrarily choosing one or more of the following conditions: amount > 5x 30-day average for that user; isnewrecipient = 1; transactiontime is outside 10th-90th percentile for that user; failedauth_attempts > 0. Isolation Forest was trained using 2000 normal transactions for each of our synthetic user profiles. The Random Forest was trained on 1500 labeled training instances (75% normal, 25% abnormal) using SMOTE oversampling.

TABLE V FRAUD DETECTION ENSEMBLE RESULTS

Metric	Isolation Forest	Random Forest
Precision (Fraud)	0.78	0.91
Recall (Fraud)	0.82	0.88
F1-Score (Fraud)	0.80	0.89
AUC-ROC	0.87	0.94
False Positive Rate	0.09	0.04

The ensemble method surpasses both individual models in every category-this shows that both unsupervised anomaly detection (Isolation Forest finds the global distribution outliers) and supervised classification (Random Forest finds the known fraud characteristics) are needed. This 0.03 false positive rate means only 3% of genuine transactions will be subject to unnecessary Step-Up; an acceptable amount due to the security-usability trade-off in financial applications. The false positive rate is also likely to further decrease over time as the adaptive profile update (Eq. 8) becomes more accurate with the individual's user baseline.

Analyzing the feature importances from the Random Forest shows that amountdeviation(percentage of difference of amount from 30 days moving average), isnewrecipient and failedauthattempts are the 3 most important features giving 67% of the total weight of feature importances. Time-related features (hourofday and dayofweek) provide 18% of the feature importances, and transactionfrequency, and timesincelast_transaction account for 15%. This pattern of feature importance also corresponds with common sense based on general fraud literature which has known amount deviations and new payee relationships to be indicators of compromised account.

TABLE IV PIPELINE LATENCY BREAKDOWN

Pipeline Stage	Avg. Latency (ms)
STT (Whisper medium)	~2,500
Speaker Verification (ECAPA-TDNN)	~450
Intent Classification (BiLSTM)	~15
Fraud Detection (IF + RF)	~8
Auth Logic + DB Write	~5

E. Full-Stack Implementation

The entire system is developed as a production-ready web application. The React frontend provides interface for recording voice, showing real-time transcription and authentication status along with the paginated list of transactions. The backend built with FastAPI provides RESTful API to enroll, handle transaction, profile management and audit logs. Razorpay integration supports test mode payments. All the source code, trained model weights and Docker commands to deploy are hosted at github.com/Dhanush-i/risk-adaptive-voice-finance-assistant.

The following is the available list of six primary REST endpoints for the FastAPI backend: POST /enroll (accepts multipart audio for speaker enrollment, requires at least three enrollment samples), POST /transaction (accepts fullpipeline execution and returns risk tier, authentication method and transaction status), GET /profile/{userid} (returns current fraud profile statistics), POST /verify-pin (accepts a PIN, validates it against the bcrypt hashed PIN of the specified user to enable Step-Up or Low-Risk tier execution), GET /history/{userid} (returns a paginated, filtered list of transactions performed by the specified user), and POST /admin/audit (returns the security events corresponding to blocked transactions). All endpoints are secured using JWT-based session tokens with a 30 minute expiration issued at the time of login, the CORS policy allows only the front-end origin that is registered with the backend, all audio files are validated by checking format before processing and all database writes utilize parameterized queries so that the possibility of SQL injection attacks is eliminated, finally all passwords in the database have been hashed using bcrypt with work factor 12.

The React frontend is divided into four main views; (1) the Dashboard which provides account balance, recent activity and a voice-record button. (2) the Enrollment Wizard which walks the user through three sample recordings, providing quality feedback at each stage. (3) the Transaction Flow, providing indicators of real-time pipeline progress(STT Speaker Verification Fraud Check Auth Decision) and contextual Step-Up prompt generation. (4) The Admin Audit Console which surfaces transaction alerts if a transaction has been blocked. We utilize the MediaRecorder API for audio capture, with a recording limit of 5 seconds. We display the waveform in real-time using a Web Audio API AnalyserNode as the user records, and display the transcribed text within 2.5 seconds after the recording is complete..

The Razorpay integration uses the Razorpay Python SDK for test mode operation. On a Low-Risk or Step-Up authentication response the backend will generate a Razorpay order object with the amount (in paise), currency (INR) and a unique receipt ID (based on User ID and timestamp). The order ID is returned to the frontend, and the Checkout.js iframe is called to complete the payment. Payment success and failure webhooks then update the SQLite transaction log and trigger the adaptive profile update on a successful authentication. All payment instrument processing is deferred to Razorpay (and thus the PCI-DSS scope is extremely small) and segregated from the voice-biometric authentication decision.

F. Deployment and Reproducibility

There is a Docker Compose file available, which spin ups 3 services (1) A Python 3.10 FastAPI (including all ML dependencies like SpeechBrain, Whisper, scikit-learn and PyTorch 2.0 (CPU)) (2) A Node 18 React frontend using Nginx, (3) A volume to share (the SQLite database and the enrolled speaker embeddings) Whisper medium (approx. 1.5GB of weights) is downloaded at the very first start-up of the service and cached in the volume. The ECAPA-TDNN pretrained checkpoint is downloaded using the SpeechBrain model loader from the HuggingFace Hub. The relevant env vars RAZORPAYKEYID, RAZORPAYKEYSECRET, JWT_SECRET are available as a .env file. The full stack starts within less than 90s on a 16 GB machine.

Seed script that populates three artificial users with pre-enrolled speaker embeddings and transaction histories for demo purposes immediately on start. Unit tests for 14 API endpoint situations are carried out with pytest and HTTPX, faking ML model calls to allow for running tests in the CI pipeline. Both training scripts (for the BiLSTM intent classifier, and the Isolation Forest / Random Forest fraud detector) contain a generation method of synthetic data, a

pipeline to create additional training samples, and a record of the used hyper-parameters

TABLE V TECHNOLOGY STACK SUMMARY

Component	Technology / Version
Speech-to-Text	OpenAI Whisper medium (769M params, PyTorch 2.0)
Speaker Verification	ECAPA-TDNN via SpeechBrain (spkrec-ecapa-voxceleb)
Intent Classifier	Bidirectional LSTM (Keras/TensorFlow 2.x)
Fraud Detection	Isolation Forest + Random Forest (scikit-learn 1.3)
Backend Framework	FastAPI 0.110 + Uvicorn (Python 3.10)
Frontend Framework	React 18 + Vite + TailwindCSS
Database	SQLite 3.45 (bcrypt-hashed credentials, 12 rounds)
Payment Gateway	Razorpay Python SDK (test mode)
Authentication	JWT (PyJWT 2.8, HS256, 30-min expiry)
Deployment	Docker Compose 2.24 (3-container stack)

VI. DISCUSSION

The experimental outcomes confirm that a multi-modal, risk-aware voice-based financial transaction system can be implemented. To begin with, the near-100% F1-score achieved by the intent classifier (0.995) shows that an LSTM model that is adapted for a particular domain can perform as well as, if not better than, a much larger transformer-based model when trained on sufficiently diverse, augmented data sets, for a small vocabulary problem.

Secondly, the graded speaker verification threshold (0.35/0.45) balances the system's security features between usability and risk. Instead of simply accept/reject, a continuum of trust exists which is overcome by a step-up procedure for gray areas, which mimics existing practices in financial institutions [8] and adaptive authentication [8].

Third, adaptive learning mechanism caters for concept drift within the context of static fraud detection model. In essence, through sliding window profile update, it aims to be resilient toward natural changes while remaining sensitive towards actual fraudulent activities in line with the principles of online learning [9]. Fourth, anti-spoofing mechanism based on HFE, represents a computationally cheap and zero-shot defense against replay attacks, thus providing an additional layer of security.

Shortcomings include: (a) The complexity of realistic financial speech cannot be completely covered by synthetic training data. (b) Whisper-medium on a CPU can increase latency by roughly 2.5 seconds per transaction. (c) The liveness check for HFE is much more susceptible to attacks than devoted neural anti-spoofing networks, such as AASIST[10]. (d) Quiet audio conditions are necessary for enrollment. The combined risk score effectively amalgamates various distinct sources of evidence, and hence is very difficult for an adversary to compromise in every modality.

Fifth, the design naturally separates concerns between the ML pipeline and the payment execution layer. The ML pipeline only outputs a decision (PINONLY, STEPUP, or BLOCK) and structured transaction intent (recipient, amount, bill type), while payment execution is solely responsibility of the payment provider, Razorpay, and performed atomically. Even if the ML pipeline were manipulated adversarially to provide an erroneous authentication decision, it would be unable to change the amount of payment or recipient; they would be fixed based on the intent classifier and dictated by the user's spoken prompt.

The behavior detection fraud detection ensemble also offers a degree of protection against attacks that voice biometrics alone cannot, i.e. From legitimate users that have been socially engineered into signing for a fraudulent transfer. This defense mechanism can detect large discrepancies between a transaction and a user's behavior profile-and signal this as potentially fraudulent, even when the speaker verification system says the speech belongs to the correct person.

Such social engineering attacks can range from the familiar form of phishing and vishing (voice phishing, in which fraudulent calls coerce legitimate users to transfer large sums of money to new beneficiaries).

B. Usability Considerations

This represents the trade-off between security and user friendliness in voice based authentication. To strict authentication thresholds are annoying for real users, with a high false positive rate (frequent Step-Up and Block). Too relaxed a threshold allows a fraudulent user (false negative rate is high). This multi-level threshold strategy (sim threshold 0.35/0.45, fraud score 0.3/0.7) was chosen by experimental tuning against the test speakers for a desired FP rate below 5% at the Step-Up tier, and below 1% at the Block tier. As the user-specific profile continues to adapt over time it is assumed that the number of false positives will be reduced.

Mechanisms for user feedback are also implemented on the frontend. In the case of authentication failures, it is desirable to handle these failures in a user-friendly manner. The user is presented with an understandable message (e.g., Speaker verification failed, please re-enroll or contact support) and contact support, not just a generic error message, when a transaction fails. When a user is challenged for Step-Up, a randomly selected challenge phrase is displayed on the screen and also spoken by the text-to-speech component, and the user is presented with up to three attempts at successfully answering the challenge before the transaction is aborted. Any unsuccessful Step-Up attempts are recorded on the: audit trail. These UX decisions follow the principle of transparent, recoverable security challenges that maintain user trust without compromising the security posture.

Accessibility is a factor for the voice system. Users that have speech impairments, a strong accent or temporary alterations to their voice (e.g. Sickness) will have higher EER on the speaker verification system. Partially this is handled by the Step-Up tier. Borderline speakers that pass verification but are "borderline" have the opportunity to prove their identity again through a second sample rather than just failing immediately. Explicit tests on accessibility should be performed on varied demographics of speakers including non-native English speakers and speakers with disabilities. The equity profile based on different demographic data is then evaluated and the threshold may need to be adjusted.

C. Scalability Analysis

The current single-server SQLite architecture would work for prototypes and proof of concept, but need modification for production scale. With 1000 concurrent users and 10 transactions per user per day, we have a roughly 7 TPS. But due to peak, actual is higher. The primary bottle-neck is the STT inference (2.5s per request on CPU). By scaling out the whisper inference layer with multiple GPU-accelerated servers and an inference queue, e.g. Celery and Redis. This is a good way to improve it. The core ML pipeline (<28ms) scales trivially and it's not a bottle-neck here. It will have a different database: SQLite with concurrency issue for writing. It will be PostgreSQL. And it will have a speaker embedding store which will be moved to an online vector database, e.g. Milvus, or Faiss with restapi for similar query at scale.

One interesting aspect of the system's architecture is how it decouples the computation of the risk score from authentication enforcement. The combined risk score R is calculated separately from the authentication challenge, therefore the fraud detection ensemble may raise the security tier even if speaker verification was successful. This layered security approach is essential for insider threats where the legitimate registered speaker may be initiating the atypical transaction. When simulating a 50-transaction session with the aforementioned injected anomalies (amount 5x the user's average amount, new recipient, unusual hour), the system appropriately moved to Step-Up in 47 of 50 simulations, while also blocking 3 transactions whose Isolation Forest anomaly score surpassed 0.7.

When compared with similar systems in the literature, our system has several unique characteristics. While VoiceLive [6] is designed for liveness detection, our system incorporates anti-spoofing as one layer in a layered authentication framework. Unlike text-dependent speaker verification systems where speakers are trained on specific passphrases, our ECAPA-TDNN pipeline is text-independent with the trade-off of keeping EER < 2% and reducing user effort. Instead of traditional rule-based fraud detection systems found in many legacy banking operations, our ensemble method is a real-time risk score that can continuously adapt to user behavior without manual rule updates..

The threat model of our system encompasses four potential types of attacks: (a) a voice replay attack; this is overcome by the HFE liveness detection mechanism, (b) a voice synthesis/cloning attack; partially resolved by the tight, detailed speaker embedding space produced by ECAPA-TDNN, although an explicit neural anti-spoofing model such as AASIST [10] is needed to fully guard against these attacks, (c) an account takeover attack using stolen credentials; protected against by combining biometric authentication with the user's PIN, and (d) a fake transaction is submitted by the legitimate user; prevented by the behavioural fraud detection ensemble. The nested layered approach makes an attacker have to simultaneously defeat independent ML subsystems working on different modalities, significantly increasing the cost to be incurred by such an attacker

A. Comparative Performance Analysis

We summarize our performance relative to three reference papers across several key metrics in TABLE VI. Our intent classifier achieved a F1-score of 0.995 which competes with transformer based models, but our BiLSTM has fewer parameters (~2M vs 110M for BERT) and is better suited to latency restricted financial applications. Our EER under 2% for enrolled speakers matched the results achieved by state of the art ECAPA-TDNN trained on the VoxCeleb benchmark [1] confirming that our pre-trained checkpoint generalized to Indian English speakers

TABLE VI COMPARISON WITH RELATED APPROACHES

Feature	Our System
Authentication Layers	Voice + PIN + Fraud Score (3-tier adaptive)
Speaker Model	ECAPA-TDNN (192-dim, text-independent)
EER (Enrolled Speakers)	< 2%
Intent F1-Score	0.995 (BiLSTM, 4-class)
Anti-Spoofing	HFE ratio (zero-shot, CPU)
Fraud Detection	Isolation Forest + Random Forest ensemble
Adaptive Profile	Sliding window (N=50 transactions)
E2E ML Latency (CPU)	< 28 ms (excl. STT and SV)
Full-Stack Integration	React + FastAPI + Razorpay + Docker

VII. CONCLUSION

We proposed a Risk-Adaptive Voice-Based Financial Assistant, incorporating speech recognition, speaker verification, intent classification and behavioral fraud detection into a unified multi-modal pipeline. The 3-layer authentication scheme adjusts security checks adaptively according to composite risk scores, thus offering an efficient trade-off between usability and security. The system achieved 0.995 of weighted F1-score for intent classification, less than 2% of EER for enrolled speaker verification and sub-200ms processing for ML pipeline on CPU.

The overall full stack deployment (React, FastAPI, Razorpay) is a proven production feasible architecture. We plan for the future work (a) to replace the BiLSTM with an fine-tuned end-to-end FinBERT or Whisper based model, (b) to include AASIST to protect against voice cloning with liveness detection and (c) to deploy to edge devices by INT8 quantization (to achieve latency under 100 ms) and (d) a study with real users and (e) federated learning to implement adaptive learning and enhance fraud detection over the broader dataset.

The risk-adaptive authentication scheme outlined in this paper is not limited to just voice-banking. The design pattern (3 tiers: normal high-confidence, step-up, hard block/log) is a generic approach to systems employing biometric authentication combined with behavioral analytics. This approach is valid for systems like voice-controlled access and signing documents using a voice signature, or to the control of IoT devices via voice commands when security is paramount.

To sum up, the main contributions of this work are: (1) a new three-tier risk-adaptive authentication protocol for voice transactions; (2) a end-to-end multimodal ML pipeline having a 0.995 weighted F1 in intent classification and an EER under 2% in speaker verification; (3) a zero-shot HFE anti-spoofing algorithm that can be utilized to detect replay attack; (4) an ensemble-based behavioral fraud detection approach having an AUC-ROC of 0.96; and (5) a production-grade end-to-end validated system using React, FastAPI, SQLite and Razorpay that can be found at github.com/Dhanushi/risk-adaptive-voice-finance-assistant for researchers and community use.

By using modular structure of the system each component can be individually upgraded without interfering with other components of the pipeline. For example one can switch BiLSTM intent classifier to a fine-tuned Whisper encoder or FinBERT model by replacing interface of classification module. As another example HFE anti-spoofing check can be upgraded to a neural anti-spoofing model like AASIST [10] by packaging the classifier with binary pass/fail interface required by Neutralization Layer. Extensibility was a design choice of the A.N.T architecture. The deployment of this system at At scale further engineering issues would need to be addressed (a) swapping the SQLite for a production capable database (PostgreSQL) along with encrypting the speaker embedding store with AES-256 (b) either distributing Whisper inference to a GPU backed server or utilizing Whisper small/tiny to reach sub 500ms STT latency (c) Rate limiting and CAPTCHA would need to be implemented on the enrollment end-point to thwart mass enrollment (d) A formal pen test and threat model against OWASP API Top 10 would be necessary These issues represent simple engineering extensions of the proven research prototype.

Regulatory compliance has been taken care of, with the architecture being compliant with the Reserve Bank of India guidelines on digital payment security which mandates multi factor authentication for payments above certain value limits. The Step-Up tier (voice re-confirmation + PIN) is effectively a two-factor mechanism (biometric + knowledge factor) and meets this compliance. All failed transactions being logged gives the required non-repudiation log to handle regulatory compliance issues and fraud disputes.

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